

API INTEGRATION DOCUMENTATION

# Digiman+ Integration Interface Specification untuk MKP

Dokumen spesifikasi teknis integrasi dari atau ke platform Digiman+ (dibuat khusus untuk MKP).

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| Versi | Tanggal Rilis | Disiapkan untuk   | Disiapkan oleh       |
|-------|---------------|-------------------|----------------------|
| 2.0   | 06 Jul 2026   | Tim Integrasi MKP | Tim Digiman+ (BTECH) |

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# 1. Pendahuluan

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Dokumen ini menjelaskan framework integrasi antara Digiman+ dan sistem eksternal (ERP seperti SAP/Oracle, atau third-party lain). Tujuan dokumen:

- Menjelaskan mekanisme integrasi
- Mendefinisikan struktur data yang dibutuhkan
- Menjelaskan spesifikasi API
- Menjabarkan kebutuhan teknis & konfigurasi dari sisi client
- Memastikan kesiapan sebelum implementasi integrasi

Ditujukan untuk membantu client memahami cara kerja integrasi, mengidentifikasi field wajib & kebutuhan konfigurasi, serta menemukan gap informasi/prasyarat teknis sebelum mulai integrasi.

## 2. Ikhtisar Integrasi

### 2.1 Kapabilitas Integrasi

Digiman+ mendukung pertukaran data dengan sistem eksternal melalui HTTP:

- **Outbound Integration:** data dari Digiman+ → sistem eksternal
- **Inbound Integration:** data dari sistem eksternal → Digiman+

### 2.2 Teknologi Integrasi

- **Tipe Integrasi:** REST API
- **Format Data:** JSON
- **Protokol Komunikasi:** HTTPS

### 2.3 Autentikasi & Kontrol Akses

1. **API Key Authentication** — semua request wajib menyertakan API Key yang valid di header X-API-Key. Request tanpa API Key valid akan ditolak. API Key di-provide oleh tim Digiman+ (client tidak generate sendiri).
2. **IP Whitelisting** — akses API dibatasi hanya untuk IP client yang sudah terdaftar. Client wajib menyediakan:
  - Public IP address(es) yang dipakai untuk API calls
  - IP per environment (UAT dan Production)

Whitelisting dilakukan oleh tim Digiman+ berdasarkan IP yang disediakan client tersebut. Request dari IP yang tidak di-whitelist akan diblokir di level network.

### 2.4 Base URL per Environment

Nilai base\_url yang dipakai di seluruh endpoint Inbound API pada dokumen ini (section 6) mengacu ke tabel berikut:

| ENVIRONMENT | BASE URL                                                                       |
|-------------|--------------------------------------------------------------------------------|
| Dev         | https://cst-shared-apm-dev-001.azure-api.net/integration/external-integration/ |

Base URL di atas sudah termasuk path integration/external-integration/, jadi endpoint tiap API (mis. .../employee, .../asset) tinggal ditambahkan langsung tanpa mengulang path tersebut.

### 3. Terminologi

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| ISTILAH              | DEFINISI                                                                                                                       |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Digiman+             | SaaS platform dari BTECH untuk manajemen aktivitas asset maintenance.                                                          |
| External System      | Sistem pihak ketiga mana pun yang terintegrasi dengan Digiman+.                                                                |
| Integration          | Mekanisme teknis pertukaran data terstruktur antara Digiman+ dan sistem eksternal via REST API over HTTPS dengan payload JSON. |
| Inbound Integration  | Data flow yang berasal dari sistem eksternal dan diproses oleh Digiman+.                                                       |
| Outbound Integration | Data flow yang berasal dari Digiman+ dan dikirim ke sistem eksternal.                                                          |
| API Key              | Secret key unik yang diberikan ke sistem eksternal untuk otentikasi request ke Digiman+ API.                                   |
| IP Whitelisting      | Kontrol keamanan level network yang membatasi akses API hanya ke IP client yang sudah disetujui.                               |

## 4. Arsitektur

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Alur integrasi: **Digiman+** ⇌ **Client Middleware** ⇌ **Client System (SAP / Oracle / dll.)**

- **Inbound** (Client → Digiman+, via middleware): Employee, Asset/Equipment, Asset/Equipment Status, Material, Maintenance Order Backlog, Notification
- **Outbound** (Digiman+ → Client, via middleware): Complete MO Inspection, Close & Delete MO Inspection, Backlog Execution, Create Maintenance Order

## 5. Spesifikasi API Outbound

Outbound = Digiman+ memanggil API milik client.

### 5.1 Backlog Execution

Dipanggil saat backlog dieksekusi oleh mekanik.

**POST** {Backlog Execution Client's API}

**Header:** Content-Type: application/json

**Request**

```
{
  "MONumber": [
    "1104499011",
    "1104499012",
    "1104499013",
    "1104499014"
  ]
}
```

**Expected Response**

```
[
  {
    "MONumber": "1104499011",
    "SAPStatus": "1",
    "SAPText": "Order has been CLOSED|Save log history success",
    "ModifiedUtcDate": "2026-01-15T14:44:37.8770544+00:00"
  }
]
```

| FIELD               | TYPE                  | NULLABLE | DESKRIPSI                                      |
|---------------------|-----------------------|----------|------------------------------------------------|
| MONumber (request)  | Array list of varchar | No       | List Maintenance Order number yang dieksekusi  |
| MONumber (response) | varchar               | No       | Maintenance Order number — satu per item hasil |
| SAPStatus           | varchar               | No       | Status integrasi: 1 = Berhasil, 2 = Gagal      |
| SAPText             | varchar               | No       | Pesan dari integrasi SAP                       |
| ModifiedUtcDate     | Datetime (ISO 8601)   | No       | Tanggal dan waktu                              |

### 5.2 Create Maintenance Order

Dipanggil saat order di-approve.

**POST** {Create MO Client's API}

**Header:** Content-Type: application/json

**Request**

```
[
  {
    "EMolNumber": "IFD-2602001584",
    "SupervisorId": 1100800565,
    "Body": [
      {
        "Id": 4807,
        "SiteId": 2009,
        "PoolingId": 4628,
        "MOType": "MT01",
        "MODescription": "P1-Flushing AC Function Internal ",
        "PMActType": "IH7",
        "Equipment": "DAVV40022",
        "BasicStartDate": "2026-03-11T07:54:21.293",
        "MaterialNumber": "1740467",
        "MaterialQuantity": "2.00",
        "Batch": "",
        "MOCreatedBy": "20090BH.supervisor01@protonmail.com",
        "AttachmentUrl": "https://digiman-dev.bukittechnology.com/order-list/material-order?workOrderId=20356&assetNumbe",
        "SLoc": "WH01",
        "MoDescriptionLong": "Model Unit : A40-G\nCode Unit : DAVV40022\nDamage : Air Leaks-Internal\nInspector : 20090B"
      }
    ]
  }
]
```

**Expected Response**

```
[
  {
    "MONumber": "1105239395",
    "SAPStatus": "1",
    "SAPText": "Success Create MO: 1105239395 and log history",
    "ModifiedUtcDate": "2026-01-15T14:44:37.8770544+00:00"
  }
]
```



| FIELD                      | TYPE                | NULLABLE | DESKRIPSI                                                                  |
|----------------------------|---------------------|----------|----------------------------------------------------------------------------|
| EMolNumber                 | varchar             | No       | Nomor electronic mechanic order                                            |
| SupervisorId               | numeric             | No       | Nomor identitas supervisor                                                 |
| Body                       | array               | No       | List of MO detail records yang akan dibuat                                 |
| Body.Id                    | integer             | No       | Id sinkronisasi                                                            |
| Body.SiteId                | numeric             | No       | Id site                                                                    |
| Body.PoolingId             | integer             | No       | Id pooling                                                                 |
| Body.MOType                | varchar             | No       | Tipe maintenance/work order. Bervariasi per client, butuh konfigurasi.     |
| Body.MODescription         | varchar             | No       | Deskripsi Maintenance Order / Work Order                                   |
| Body.PMActType             | varchar             | No       | Tipe Plant Maintenance Activity. Bervariasi per client, butuh konfigurasi. |
| Body.Equipment             | varchar             | No       | Nomor equipment                                                            |
| Body.BasicStartDate        | Datetime (ISO 8601) | No       | Tanggal mulai                                                              |
| Body.MaterialNumber        | varchar             | No       | Nomor material                                                             |
| Body.MaterialQuantity      | varchar             | No       | Jumlah material yang dipesan                                               |
| Body.Batch                 | varchar             | Yes      | Batch material                                                             |
| Body.MOCreatedBy           | varchar             | No       | User code pembuat order                                                    |
| Body.AttachmentUrl         | varchar             | No       | Bukti/evidence temuan                                                      |
| Body.SLoc                  | varchar             | No       | Storage location material                                                  |
| Body.MoDescriptionLong     | varchar             | No       | Deskripsi panjang MO/WO                                                    |
| MONumber (response)        | varchar             | No       | MO number yang digenerate ERP                                              |
| SAPStatus (response)       | varchar             | No       | 1 = Berhasil, 2 = Gagal                                                    |
| SAPText (response)         | varchar             | No       | Pesan dari integrasi SAP                                                   |
| ModifiedUtcDate (response) | Datetime (ISO 8601) | No       | Tanggal dan waktu                                                          |

## 6. Spesifikasi API Inbound

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Inbound = client memanggil API Digiman+.

### 6.1 Employee

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Upsert employee — matched by `employees.userCode`; kalau sudah ada → update, kalau belum → create. Mendukung single & bulk.

**POST** `base_url/employee`

#### Request

```
{
  "employees": [
    {
      "userCode": "mkpinspector01@protonmail.com",
      "fullName": "mkpinspector01",
      "employeeId": "0000895",
      "positionCode": "C01",
      "positionName": "Supervisor",
      "levelCode": "3A",
      "levelName": "3A",
      "siteCode": "1001",
      "siteName": "Sesayap",
      "departmentCode": "PLT",
      "departmentName": "Plant",
      "sectionCode": "S001",
      "sectionName": "Hauler",
      "isActive": true
    }
  ]
}
```

| FIELD                    | TYPE         | NULLABLE | DESKRIPSI                                                                                                  |
|--------------------------|--------------|----------|------------------------------------------------------------------------------------------------------------|
| employees                | array        | No       | Daftar record employee (mendukung pengiriman bulk)                                                         |
| employees.userCode       | varchar(64)  | No       | Identifier unik user; harus unik                                                                           |
| employees.fullName       | varchar(150) | No       | Nama lengkap employee                                                                                      |
| employees.employeeId     | varchar(50)  | No       | ID Employee (NIK)                                                                                          |
| employees.positionCode   | varchar(64)  | Yes      | Kode posisi/jabatan employee                                                                               |
| employees.positionName   | varchar(64)  | Yes      | Nama posisi/jabatan employee                                                                               |
| employees.levelCode      | varchar(64)  | Yes      | Kode level employee                                                                                        |
| employees.levelName      | varchar(64)  | Yes      | Nama level employee                                                                                        |
| employees.siteCode       | varchar(64)  | No       | Site tempat employee ditugaskan. Jika kode belum terdaftar, akan otomatis di-insert ke master data.        |
| employees.siteName       | varchar(64)  | No       | Nama site                                                                                                  |
| employees.departmentCode | varchar(64)  | No       | Department identifier (referensi saja). Jika kode belum terdaftar, akan otomatis di-insert ke master data. |
| employees.departmentName | varchar(64)  | No       | Nama department                                                                                            |
| employees.sectionCode    | varchar(64)  | No       | Identifier Section user. Jika kode belum terdaftar, akan otomatis di-insert ke master data.                |
| employees.sectionName    | varchar(64)  | No       | Nama section                                                                                               |
| employees.isActive       | boolean      | No       | true = aktif, false = tidak aktif                                                                          |

## Response

### HTTP 200 – Success

```
{
  "status": "success",
  "totalData": 1,
  "successCount": 1,
  "failedCount": 0,
  "results": [
    { "userCode": "mkpinspector01@protonmail.com", "status": "success" }
  ]
}
```

### HTTP 200 – Partial Success

```
{
  "status": "success",
  "totalData": 2,
  "successCount": 1,
  "failedCount": 1,
  "results": [
    { "userCode": "mkpinspector01@protonmail.com", "status": "success" },
    { "userCode": "mkpinspector02@protonmail.com", "status": "failed", "errorCode": "ERR-0017" }
  ]
}
```

## HTTP 400

```
{
  "status": "error",
  "totalData": 1,
  "successCount": 0,
  "failedCount": 1,
  "results": [
    { "userCode": "mkpinspector01@protonmail.com", "status": "failed", "errorCode": "ERR-0017" }
  ]
}
```

**HTTP 403:** { "errorCode": "ERR-0006" }   **HTTP 500:** { "errorCode": "ERR-0007" }

## 6.2 Asset & Asset Status

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### 6.2.1 Asset

Upsert asset — matched by `assets.assetNumber`. Mendukung single & bulk.

**POST** `base_url/asset`

#### Request

```
{
  "assets": [
    {
      "assetNumber": "EXC00031",
      "siteCode": "1001",
      "siteName": "Sesayap",
      "sectionTypeCode": "OBL",
      "sectionTypeName": "Loader",
      "assetCategoryCode": "A",
      "assetTypeCode": "EXC",
      "assetTypeName": "Excavator",
      "assetBrandCode": "KOM",
      "assetBrandName": "KOMATSU",
      "assetModelCode": "PC2000-8",
      "assetModelName": "PC2000-8",
      "assetClassName": "EXA 200 TON CLASS",
      "assetVariantName": "EXA 200 TON CLASS",
      "isActive": true
    }
  ]
}
```

| FIELD                     | TYPE                 | NULLABLE | DESKRIPSI                                                                                            | TABEL SAP                   | CATATAN                           |
|---------------------------|----------------------|----------|------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------|
| assets                    | array                | No       | Daftar record asset (mendukung pengiriman bulk)                                                      |                             |                                   |
| assets.assetNumber        | varchar(200)         | No       | Identifier unik asset                                                                                | EQUI-EQUNR                  |                                   |
| assets.siteCode           | varchar(64)          | No       | Site tempat equipment ditugaskan. Jika kode belum terdaftar, akan otomatis di-insert ke master data. | ILOA-SWERK                  |                                   |
| assets.siteName           | varchar(64)          | No       | Nama site                                                                                            |                             |                                   |
| assets.sectionTypeCode    | varchar(64)          | No       | Identifier section/area equipment. Tidak diproses jika belum terdaftar.                              | ILOA-BEBER                  | contoh: 201                       |
| assets.sectionTypeName    | varchar(512)         | No       | Nama section/area (pasangan sectionTypeCode)                                                         | T357-FING                   | contoh: Hauler                    |
| assets.assetCategoryCode  | varchar(64)          | Yes      | Kode kategori asset. Tidak diproses jika belum terdaftar.                                            | EQUI-EQTYP                  | contoh: H, M                      |
| assets.assetCategoryName  | varchar              | Yes      | Nama category (pasangan assetCategoryCode — keduanya wajib atau tidak sama sekali)                   | T370T-TYPTX                 | contoh: Heavy Equipment, Machines |
| assets.assetTypeCode      | varchar(64)          | Yes      | Kode tipe asset (pasangan assetTypeName)                                                             | EQUI-EQART                  | contoh: 01, 02                    |
| assets.assetTypeName      | varchar (trunc. 200) | Yes      | Nama tipe asset. Tidak diproses jika belum terdaftar.                                                | T370K-EARTX                 | contoh: Excavator                 |
| assets.assetBrandCode     | varchar(64)          | No       | Kode brand asset (pasangan assetBrandName). Wajib diisi.                                             | AUSP-ATWRT / EQUI-HERST     | contoh: KOMATSU                   |
| assets.assetBrandName     | varchar(128)         | No       | Nama brand asset. Tidak diproses jika belum terdaftar.                                               | AUSP-ATWRT / EQUI-HERST     | contoh: KOMATSU                   |
| assets.assetModelCode     | varchar(64)          | No       | Kode model asset (pasangan assetModelName). Wajib diisi.                                             | AUSP-ATWRT / EQUI-TYPBZ     | contoh: PC2000-8                  |
| assets.assetModelName     | varchar(50)          | No       | Nama model asset. Tidak diproses jika belum terdaftar.                                               | AUSP-ATWRT / EQUI-TYPBZ     | contoh: PC2000-8                  |
| assets.assetClassCode     | varchar(64)          | Yes      | Kode class asset (pasangan assetClassName)                                                           | TBD — konfirmasi tim SAP    | <b>Tidak Pakai</b>                |
| assets.assetClassName     | varchar(128)         | Yes      | Nama class asset. Tidak diproses jika belum terdaftar.                                               | AUSP-ATWRT                  | <b>Tidak Pakai</b>                |
| assets.assetVariantCode   | varchar(64)          | Yes      | Kode variant asset (pasangan assetVariantName)                                                       |                             | <b>Tidak dipakai</b>              |
| assets.assetVariantName   | varchar (trunc. 200) | Yes      | Nama variant asset. Tidak diproses jika belum terdaftar.                                             | AUSP-ATWRT                  | → <b>tidak dipakai</b>            |
| assets.assetOwnershipCode | varchar(64)          | Yes      | Kode kepemilikan asset                                                                               | TBD — konfirmasi tim SAP    |                                   |
| assets.isActive           | boolean              | Yes      | Opsional, default true bila diomit                                                                   | Dikirim dari SAP true/false | INAC atau DLFL = false            |

|  |  |  |  |                                 |  |
|--|--|--|--|---------------------------------|--|
|  |  |  |  | berdasarkan status<br>equipment |  |
|--|--|--|--|---------------------------------|--|

**Catatan:** assetClassCode/assetClassName dan assetVariantCode/assetVariantName ditandai **Tidak Pakai** khusus untuk konteks MKP (SAP MKP tidak menggunakan konsep class/variant ini). Field-nya tetap Nullable=Yes secara skema, jadi aman baik dikirim maupun tidak — contoh payload di atas menampilkan value hanya untuk ilustrasi format, bukan berarti field ini aktif dipakai MKP.

**Response:** sama pola dengan Employee API (assetNumber sebagai key), HTTP 200 Success/Partial Success, 400, 403 (ERR-0006), 500 (ERR-0007).

6.2.2 Asset Status

Registrasi status operasional equipment di Digiman+, sebagai historical log kondisi unit.

Client hanya perlu mengirim status **terakhir (last status)** setiap kali terjadi perubahan status — tidak perlu mengirim ulang seluruh histori status equipment di setiap request.

**POST** base\_url/asset-status

Request

```
{
  "assetStatus": [
    {
      "trDate": "2024-10-14T00:00:00+07:00",
      "assetNumber": "BDCT10003",
      "status": "IN OPERATION"
    }
  ]
}
```

| FIELD                   | TYPE                | NULLABLE | DESKRIPSI                                                      | TABEL SAP   | CATATAN                                                             |
|-------------------------|---------------------|----------|----------------------------------------------------------------|-------------|---------------------------------------------------------------------|
| assetStatus             | array               | No       | Kumpulan record status asset                                   |             |                                                                     |
| assetStatus.trDate      | datetime (ISO 8601) | No       | Tanggal/waktu efektif status di source system, dengan timezone | JCDS-UPDATE |                                                                     |
| assetStatus.assetNumber | varchar(50)         | No       | Nomor asset unik. Tidak diproses jika belum terdaftar.         | EQUI-EQUNR  |                                                                     |
| assetStatus.status      | varchar(20)         | No       | Status: 1. IN OPERATION, 2. IDLE, 3. RETIRED                   | JEST-STAT   | EQUI-OBJNR = JEST-OBJNR where JEST-INACT = NULL And JEST-STAT = 'I' |

Response

- HTTP 200 – Success:** {"status":"success","totalData":1,"successCount":1,"failedCount":0,"results":[{"assetNumber":"BDCT10003","trDate":"2024-10-14T00:00:00+07:00"}]}

- **HTTP 200 – Partial Success:** hasil per item, item gagal punya errorCode (contoh ERR-0014)
- **HTTP 400:** status: "error" dengan errorCode per item
- **HTTP 403:** ERR-0006    **HTTP 500:** ERR-0007

## 6.3 Material

---

### 6.3.1 Material

Upsert material — matched by `materials.materialNumber`. Mendukung single & bulk.

**POST**    `base_url/material`

#### Request

```
{
  "materials": [
    {
      "materialNumber": "01-085785-9",
      "materialName": "SLEEVE,SPECTRUM ADJUSTMENT,BR",
      "batchCode": "NEW",
      "siteCode": "1001",
      "sectionTypeCode": "OBH",
      "uomCode": "EA",
      "assetModelCode": "740C",
      "storageLocation": "WHA1",
      "currency": "USD",
      "map": 14.32,
      "mapLocalCurr": 234217.92,
      "stock": 25,
      "ranking": "E",
      "isActive": true
    }
  ]
}
```



| FIELD                     | TYPE         | NULLABLE | DESKRIPSI                                                         | TABEL SAP    | CATATAN                                                                                                                                                                               |
|---------------------------|--------------|----------|-------------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| materials                 | array        | No       | Daftar record material (mendukung pengiriman bulk)                |              |                                                                                                                                                                                       |
| materials.materialNumber  | varchar(200) | No       | Identifier unik material                                          | MARA-MATNR   |                                                                                                                                                                                       |
| materials.materialName    | varchar(200) | No       | Nama/deskripsi material                                           | MARA-MAKTX   |                                                                                                                                                                                       |
| materials.batchCode       | varchar(64)  | No       | Identifier batch. Contoh: NEW, REPAIRED, string kosong            | MBEW-BWTAR   | MARA-MATNR = MBEW-MATNR                                                                                                                                                               |
| materials.siteCode        | varchar(64)  | No       | Site tempat material berada. Tidak diproses jika belum terdaftar. | MBEW-BWKEY   |                                                                                                                                                                                       |
| materials.sectionTypeCode | varchar(64)  | No       | Kode section type. Tidak diproses jika belum terdaftar.           | AUSP-ATWRT   | 1) get equipment dari material master: A USP-ATINN='EQP_MOD_OEM*' and A USP-ATWRT=[material model];<br>2) get hierarki (loader/hauler): A USP-ATINN='EQP_MOD_HIR' and A USP-OBJEK=[1] |
| materials.uomCode         | varchar(64)  | Yes      | Satuan unit (UOM). Tidak diproses jika belum terdaftar.           | MARA-MEINS   |                                                                                                                                                                                       |
| materials.assetModelCode  | varchar(64)  | Yes      | Kode model asset. Tidak diproses jika belum terdaftar.            | AUSP-ATWRT   | ATINN='MODEL_UNIT'                                                                                                                                                                    |
| materials.storageLocation | varchar(64)  | Yes      | Lokasi penyimpanan material                                       | MARD-LGORT   |                                                                                                                                                                                       |
| materials.currency        | varchar(5)   | Yes      | Mata uang harga material                                          | CKMLCR-WAERS | CKMLRCURTP=10                                                                                                                                                                         |
| materials.map             | float        | Yes      | Moving average price / harga rata-rata material                   | CKMLCR-PVPRS | contoh 14.32                                                                                                                                                                          |
| materials.mapLocalCurr    | float        | Yes      | Harga dalam mata uang lokal                                       | CKMLCR-PVPRS | CKMLRCURTP=40                                                                                                                                                                         |
| materials.stock           | float        | Yes      | Jumlah stok saat ini                                              | MARD-LABST   |                                                                                                                                                                                       |
| materials.ranking         | varchar(64)  | Yes      | Ranking material (A, B, C, dst.)                                  | MARC-MAABC   |                                                                                                                                                                                       |
| materials.isActive        | boolean      | Yes      | Mengaktifkan/menonaktifkan material. Opsional, default true       | MARC-LVORM   |                                                                                                                                                                                       |

**Response:** pola sama (key materialNumber), HTTP 200/400/403(ERR-0006)/500(ERR-0007).

## 6.4 Backlog

### 6.4.1 Backlog

Registrasi **Pending Work Order** baru beserta material yang dibutuhkan, dalam satu langkah integrasi.

**POST** base\_url/backlog

**Request**

```

{
  "backlogs": [
    {
      "moNumber": "1105152450",
      "moDescription": "Replace hose compressor",
      "siteCode": "1001",
      "assetNumber": "EXCT61016",
      "orderType": "MT01",
      "assetModelCode": "6015B",
      "assetTypeCode": "EXC",
      "maintenanceCategoryCode": "BEX",
      "maintenanceCategoryName": "Breakdown Excavator",
      "object": "IDLER LH",
      "reservationNumber": "1234567890",
      "moCreatedDate": "2024-01-08T08:00:00+07:00",
      "moCreatedBy": "SAP",
      "moUpdatedDate": "2024-01-08T09:00:00+07:00",
      "moUpdatedBy": "SAP",
      "moReleaseDate": "2024-01-09T08:00:00+07:00",
      "moTecoDate": null,
      "moClosedDate": null,
      "moPlanCost": 500000.00,
      "moActualCost": 0.00,
      "moStatus": "OPEN",
      "isActive": true,
      "materials": [
        {
          "materialNumber": "MAT-001",
          "materialName": "Hydraulic Hose",
          "batch": "NEW",
          "orderQuantity": 1.00,
          "uom": "PCS",
          "storageLocation": "WH01",
          "prStatus": "CREATED",
          "prNumber": "12345",
          "poNumber": "98765",
          "orderStatus": "OPEN",
          "grStatus": "PARTIAL",
          "grQuantity": 1.00,
          "grNumber": "77889",
          "giNumber": "77889",
          "giDate": "2024-01-10T08:00:00+07:00",
          "giStatus": "PARTIAL",
          "giQuantity": 1.00,
          "estCost": 500000.00,
          "isActive": true
        }
      ]
    }
  ]
}

```

## Header Fields

| FIELD                            | TYPE         | NULLABLE | DESKRIPSI                                                                                    | TABEL SAP                | CATATAN                       |
|----------------------------------|--------------|----------|----------------------------------------------------------------------------------------------|--------------------------|-------------------------------|
| backlogs                         | array        | No       | Daftar record backlog MO (mendukung pengiriman bulk)                                         |                          |                               |
| backlogs.moNumber                | varchar(12)  | No       | Nomor MO unik dari sistem eksternal                                                          | AUFK-AUFNR               |                               |
| backlogs.moDescription           | varchar(250) | No       | Deskripsi pekerjaan maintenance                                                              | AUFK-KTEXT               |                               |
| backlogs.siteCode                | varchar(4)   | No       | Site/plant code. Numeric only 1–4 digit (mis. "1001"). Tidak diproses jika belum terdaftar.  | AUFK-WERKS               |                               |
| backlogs.assetNumber             | varchar(20)  | No       | Identifier asset/equipment terkait. Data ditolak jika equipment belum terdaftar di Digiman+. | EQUI-EQUNR               |                               |
| backlogs.orderType               | varchar(20)  | Yes      | Tipe maintenance order (mis. MT01). Tidak diproses jika belum terdaftar.                     | AUFK-AUART               |                               |
| backlogs.assetModelCode          | varchar(50)  | Yes      | Kode model asset                                                                             | TBD — konfirmasi tim SAP |                               |
| backlogs.assetTypeCode           | varchar(100) | Yes      | Kode tipe asset. Tidak diproses jika belum terdaftar.                                        | EQUI-EQART               |                               |
| backlogs.maintenanceCategoryCode | varchar(16)  | No       | Kode kategori maintenance activity (PM, BEX, dll). Tidak diproses jika belum terdaftar.      | AFIH-ILART               | AUFK-AUFNR = AFIH-AUFNR       |
| backlogs.maintenanceCategoryName | varchar(100) | Yes      | Nama kategori (pasangan maintenanceCategoryCode)                                             | TBD — konfirmasi tim SAP | → PM Activity Type            |
| backlogs.object                  | varchar(50)  | Yes      | Tipe Technical Object, Komponen Utama                                                        | EQUI-EQART               | AFIH-EQUNR = EQUI-EQUNR       |
| backlogs.reservationNumber       | varchar(19)  | Yes      | Nomor reservasi. Hanya karakter numerik.                                                     | TBD — konfirmasi tim SAP | Numeric only (^\\d+\$), Int64 |
| backlogs.moCreatedDate           | datetime     | No       | Tanggal MO dibuat di source system, ISO 8601                                                 | AUFK-ERDAT               |                               |
| backlogs.moCreatedBy             | varchar(100) | Yes      | User/system pembuat MO                                                                       | AUFK-ERNAM               |                               |
| backlogs.moUpdatedDate           | datetime     | Yes      | Tanggal MO terakhir diupdate                                                                 | AUFK-AEDAT               |                               |
| backlogs.moUpdatedBy             | varchar(100) | Yes      | User/system yang terakhir update MO                                                          | AUFK-AENAM               |                               |
| backlogs.moReleaseDate           | datetime     | Yes      | Tanggal MO dirilis untuk eksekusi                                                            | AFKO-FTRMI               | AUFK-AUFNR = AFKO-AUFNR       |

|                       |               |     |                                                           |            |                                                                            |
|-----------------------|---------------|-----|-----------------------------------------------------------|------------|----------------------------------------------------------------------------|
| backlogs.moTecoDate   | datetime      | Yes | Tanggal MO TECO (technically complete)                    | AUFK-IDAT2 |                                                                            |
| backlogs.moClosedDate | datetime      | Yes | Tanggal MO ditutup administratif                          | AUFK-IDAT3 |                                                                            |
| backlogs.moPlanCost   | decimal(18,2) | Yes | Estimasi biaya (planned cost)                             | COSP-WTG*  | AUFK-AUFNR = COSP- '*OBJNR',<br>COSP-VRGNG = 'KPPP'                        |
| backlogs.moActualCost | decimal(18,2) | Yes | Biaya aktual                                              | COSP-WTG*  | AUFK-AUFNR = COSP- '*OBJNR',<br>COSP-VRGNG = 'COIN'                        |
| backlogs.moStatus     | varchar(100)  | Yes | 1. OPEN, 2. RELEASED, 3. COMPLETED, 4. CLOSED, 5. DELETED | JEST-STAT  | AUFK-OBJNR = JEST-OBJNR<br>where JEST-INACT = NULL<br>And JEST-STAT = 'I*' |
| backlogs.isActive     | boolean       | No  | true = MO aktif, false = MO tidak aktif                   | JEST-STAT  | True = I0001, I0002, I0045, I0046, atau I0327. False = I0043               |

#### Material Array Fields (backlogs.materials)

| FIELD                     | TYPE          | NULLABLE | DESKRIPSI                                                   | TABEL SAP      | CATATAN                                                                                               |
|---------------------------|---------------|----------|-------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------|
| materials                 | array         | Yes      | List material terkait backlog                               |                |                                                                                                       |
| materials.materialNumber  | varchar(50)   | No       | Identifier material                                         | RESB-MATNR     | RESB-AUFNR = AUFK-AUFNR                                                                               |
| materials.materialName    | varchar(100)  | No       | Deskripsi material                                          | MAKT-MAKTX     | MAKT-MATNR = RESB-MATNR                                                                               |
| materials.batch           | varchar(50)   | No       | Klasifikasi usage material (NEW, REPLACE, dll; bisa kosong) | RESB-CHARG     | RESB-AUFNR = AUFK-AUFNR                                                                               |
| materials.orderQuantity   | decimal(18,2) | Yes      | Quantity yang dipesan/direservasi                           | RESB-BDMNG     | RESB-AUFNR = AUFK-AUFNR                                                                               |
| materials.uom             | varchar(50)   | No       | Satuan unit (PCS, EA, SET)                                  | RESB-MEINS     | RESB-AUFNR = AUFK-AUFNR                                                                               |
| materials.storageLocation | varchar(50)   | Yes      | Lokasi penyimpanan material                                 | RESB-LGORT     | RESB-AUFNR = AUFK-AUFNR                                                                               |
| materials.prStatus        | varchar(64)   | Yes      | Status Purchase Requisition (CREATED, APPROVED)             | EBAN-FRGKZ     | Lihat logika Release Strategy di dokumen asli                                                         |
| materials.prNumber        | varchar(19)   | Yes      | Nomor Purchase Requisition. Hanya karakter numerik.         | EBAN-BANFN     | EBAN-ZZ_AUFNR = AUFK-AUFNR                                                                            |
| materials.poNumber        | varchar(19)   | Yes      | Nomor Purchase Order. Hanya karakter numerik.               | EKKO-EBELN     | Step 1: EBAN-BANFN where EBAN-ZZ_AUFNR = AUFK-AUFNR; Step 2: EKKO-EBELN = EBAN-EBELN                  |
| materials.orderStatus     | varchar(16)   | Yes      | Status order terkait supply material                        | EKKO-FRGKE     | Lihat logika Release Strategy di dokumen asli                                                         |
| materials.grStatus        | varchar(64)   | Yes      | Status Goods Receipt                                        | EKPO-ELIKZ     | "FULL RECEIVED" jika EKPO-ELIKZ=X dan EKPO-LOEKZ=blank, else "PARTIAL"                                |
| materials.grQuantity      | decimal(18,2) | Yes      | Quantity diterima via Goods Receipt                         | MATDOC-MENGE   | MATDOC-MBLNR = materials.giNumber                                                                     |
| materials.grNumber        | varchar(40)   | Yes      | Nomor dokumen Goods Receipt (free text)                     | MATDOC-MBLNR   | MATDOC-EBELN = EKKO-EBELN where MATDOC-SMBLNR=blank and MATDOC-CANCELLED=blank and MATDOC-BWART='101' |
| materials.giNumber        | varchar(40)   | Yes      | Nomor dokumen Goods Issue (free text)                       | MATDOC-MBLNR   | MATDOC-AUFNR = AUFK-AUFNR where MATDOC-SMBLNR=blank and MATDOC-CANCELLED=blank and MATDOC-BWART='261' |
| materials.giDate          | datetime      | Yes      | Tanggal transaksi Goods Issue, ISO 8601                     | MATDOC-BUDAT + | MATDOC-MBLNR = materials.giNumber                                                                     |

|                      |               |     |                                        |              |                                                                                                 |
|----------------------|---------------|-----|----------------------------------------|--------------|-------------------------------------------------------------------------------------------------|
|                      |               |     |                                        | MATDOC-CPUTM |                                                                                                 |
| materials.giStatus   | varchar(64)   | Yes | Status Goods Issue (PARTIAL/COMPLETE)  | RESB-KZEAR   | Partial jika RESB-KZEAR != X;<br>Complete jika RESB-KZEAR = X,<br>where RESB-AUFNR = AUFK-AUFNR |
| materials.giQuantity | decimal(18,2) | Yes | Quantity yang di-issue via Goods Issue | MATDOC-MENGE | MATDOC-AUFNR = AUFK-AUFNR                                                                       |
| materials.estCost    | decimal(18,2) | Yes | Estimasi biaya material                | COSP-WTG*    | AUFK-AUFNR = COSP-'*OBJNR',<br>COSP-VRGNG = 'KPPP'                                              |
| materials.isActive   | boolean       | No  | Status aktif material record           | MARC-LVORM   |                                                                                                 |

**Response** (perhatikan key berbeda dari API lain: totalRecords/successRecords/failedRecords):

### HTTP 200 – Success

```
{"status":"success","totalRecords":1,"successRecords":1,"failedRecords":0,"results":[{"moNumber":"1105152450","status":"
```

- **HTTP 200 – Partial Success:** item gagal punya errorCode (mis. ERR-0002)
- **HTTP 400:** status:"error" dengan totalData/successCount/failedCount
- **HTTP 403:** ERR-0006   **HTTP 500:** ERR-0007

## 6.5 Breakdown Notification

### 6.5.1 Breakdown Notification

Upsert breakdown notification — matched by `breakdownNotifications.notification`; kalau sudah ada → update, kalau belum → create. Mendukung single & bulk. Hanya ada satu API (POST) untuk create maupun update.

**POST** `base_url/breakdown-notification`

#### Request

```
{
  "breakdownNotifications": [
    {
      "notification": "810002225968",
      "notificationStatus": "IN PROGRESS",
      "breakdownDescription": "Hydraulic hose leakage",
      "siteCode": "1001",
      "assetNumber": "DT777001",
      "assetModelCode": "777D",
      "breakdownType": "Unscheduled",
      "hmBreakdown": 58008.78,
      "breakdownStartDate": "2026-01-19T08:26:57+08:00",
      "breakdownEndDate": null,
      "rfuEstimateDate": "2026-01-19T17:00:00+08:00",
      "location": "TA BARAT",
      "isCloseBreakdown": false,
      "isActive": true
    }
  ]
}
```



| FIELD                                       | TYPE                | NULLABLE | DESKRIPSI                                                                                                                 | TABEL SAP                | CATATAN                                                              |
|---------------------------------------------|---------------------|----------|---------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------|
| breakdownNotifications                      | Varchar             | No       | Daftar record breakdown notification (mendukung pengiriman bulk)                                                          |                          |                                                                      |
| breakdownNotifications.notification         | Varchar(12)         | No       | Nomor notifikasi unik, wajib unik di sistem                                                                               | QMEL-QMNUM               |                                                                      |
| breakdownNotifications.notificationStatus   | Varchar(20)         | No       | 1. OPEN, 2. IN PROGRESS, 3. COMPLETED, 4. DELETED                                                                         | JEST-STAT                | QMEL-QMNUM = JEST-OBJNR where JEST-INACT = NULL And JEST-STAT = 'I*' |
| breakdownNotifications.breakdownDescription | Varchar(200)        | No       | Deskripsi kerusakan/kegagalan equipment                                                                                   | QMEL-QMTXT               |                                                                      |
| breakdownNotifications.siteCode             | Varchar(4)          | No       | Site lokasi breakdown. Tidak diproses jika belum terdaftar.                                                               | QMIH-IWERK               | QMEL-QMNUM = QMIH-QMNUM                                              |
| breakdownNotifications.assetNumber          | Varchar(25)         | No       | Identifier equipment. Data ditolak jika equipment belum terdaftar di Digiman+.                                            | EQUI-EQUNR               |                                                                      |
| breakdownNotifications.assetModelCode       | Varchar(40)         | Yes      | Kode model asset. Tidak diproses jika belum terdaftar.                                                                    | AUSP-ATWRT               | EQUI-EQUNR = AUSP.OBJEK where ATINN = EQP_MOD_OEM*                   |
| breakdownNotifications.breakdownType        | varchar (free text) | Yes      | Teks bebas — Digiman+ memetakan value server-side: "Unscheduled" → tipe Unscheduled, nilai lain default ke tipe Breakdown | ZIDMBD-BDSUBTYP          | QMEL-QMNUM = ZIDMBD-QMNUM                                            |
| breakdownNotifications.hmBreakdown          | Decimal(13,2)       | Yes      | Hour meter saat breakdown terjadi                                                                                         | ZIDMBD-HM_BD             | QMEL-QMNUM = ZIDMBD-QMNUM                                            |
| breakdownNotifications.breakdownStartDate   | Datetime            | No       | Tanggal/waktu mulai breakdown, ISO 8601 dengan timezone                                                                   | QMIH-VIQMEL + QMIH-AUZTV | QMEL-QMNUM = QMIH-QMNUM                                              |
| breakdownNotifications.breakdownEndDate     | Datetime            | Yes      | Tanggal/waktu breakdown selesai,                                                                                          | QMIH-AUSBS + QMIH-       |                                                                      |

|                                                      |              |     |                                                                                                                                     |                    |                                      |
|------------------------------------------------------|--------------|-----|-------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------------------------|
|                                                      |              |     | ISO 8601 dengan timezone. Null jika breakdown belum selesai<br>( <code>isCloseBreakdown</code> = false).                            | AUZTB              |                                      |
| <code>breakdownNotifications.rfuEstimateDate</code>  | Datetime     | Yes | Estimasi tanggal/waktu equipment kembali beroperasi (Ready For Use), ISO 8601 dengan timezone                                       | ZIDMBD-RFUESTDATE1 | QMEL-QMNUM = ZIDMBD-QMNUM            |
| <code>breakdownNotifications.location</code>         | Varchar(200) | Yes | Lokasi breakdown dalam site operasional                                                                                             | -                  | contoh: TA BARAT                     |
| <code>breakdownNotifications.isCloseBreakdown</code> | Boolean      | No  | true = breakdown sudah selesai/finish dan ditutup (closed), false = masih berlangsung/terbuka (open)                                | ZIDMBD-ISSTATBD    | QMEL-QMNUM = ZIDMBD-QMNUM            |
| <code>breakdownNotifications.isActive</code>         | boolean      | No  | Status validitas data record (bukan status breakdown). true = data valid/aktif, false = record sudah di-cancel/delete (tidak valid) | JEST-STAT          | True = I0061; False = I0060 OR I0062 |

**Response:** pola sama seperti API lain (key notification), HTTP 200/400/403(ERR-0006)/500(ERR-0007).

## 7. Glosarium — 7.1 Kode Error

| KODE     | DESKRIPSI                                                                      |
|----------|--------------------------------------------------------------------------------|
| ERR-0001 | Kode tenant yang dikirim belum terdaftar di master data sistem.                |
| ERR-0002 | Kode site yang dikirim belum terdaftar di master data sistem.                  |
| ERR-0003 | Kode department yang dikirim belum terdaftar di master data sistem.            |
| ERR-0004 | Kode section yang dikirim belum terdaftar di master data sistem.               |
| ERR-0005 | User dengan user code yang sama sudah ada di sistem.                           |
| ERR-0006 | Akses ke API ini ditolak karena permission tidak mencukupi.                    |
| ERR-0007 | Terjadi internal server error saat memproses request.                          |
| ERR-0008 | Kode tipe asset yang dikirim belum terdaftar di master data sistem.            |
| ERR-0009 | Kode class asset yang dikirim belum terdaftar di master data sistem.           |
| ERR-0010 | Kode variant asset yang dikirim belum terdaftar di master data sistem.         |
| ERR-0011 | Kode model asset yang dikirim belum terdaftar di master data sistem.           |
| ERR-0012 | Kode brand asset yang dikirim belum terdaftar di master data sistem.           |
| ERR-0013 | Kode kategori asset yang dikirim belum terdaftar di master data sistem.        |
| ERR-0014 | Identifier equipment yang dikirim belum terdaftar di sistem.                   |
| ERR-0015 | Kode batch yang dikirim belum terdaftar di master data sistem.                 |
| ERR-0016 | Kode UOM (unit of measure) yang dikirim belum terdaftar di master data sistem. |
| ERR-0017 | User code yang dikirim belum terdaftar di sistem.                              |